PHASE 1 OF 8

Brainstorming & Ideation

Identifying the problem and shaping the LegalEase idea

Project: LegalEase – AI-Powered Legal Document Generator

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1.1 Purpose of this Phase

The Brainstorming & Ideation phase is where the LegalEase project began. Its purpose was to identify a real problem worth solving, explore how Generative AI could address it, and agree on a clear, achievable project idea before any requirements, design or coding work started. The team looked at how legal documents are prepared today, where time and effort are lost, and how an AI-assisted web application could reduce that effort.

1.2 Problem Statement

Preparing legal documents such as agreements, contracts, non-disclosure agreements (NDAs), lease agreements and employment documents is largely a manual activity. Individuals, small businesses and students often start from a blank page or copy an old template, then edit party names, dates, clauses and jurisdiction details by hand. This process has several weaknesses:

- **Time-consuming:** drafting even a simple agreement requires repeated typing and rearranging of standard clauses.
- **Inconsistent structure:** documents prepared by different people follow different layouts, headings and wording.
- **Error-prone:** party names, dates or terms are easily missed or mismatched when a template is reused.
- **Formatting overhead:** after the content is written, effort is still needed to produce a clean Word or PDF document with headings and footers.

The team concluded that this problem could be supported by Generative AI: if a user supplies the essential facts, an AI model can produce a structured first draft, leaving the user to review and refine it.

1.3 Target Users

User group	Typical need	How LegalEase helps
Individuals	Rental or simple personal agreements	Generates a structured draft from a short form

User group	Typical need	How LegalEase helps
Small businesses / start-ups	NDAs, service and employment documents	Saves drafting time and keeps a consistent layout
Students and project teams	Understanding how legal drafts are structured	Shows a complete, editable example document
Administrative staff	Preparing routine documents repeatedly	Reusable workflow with export to TXT, DOCX and PDF

1.4 Idea Generation

During brainstorming the team discussed what a useful and realistic solution would look like. The ideas below were collected and agreed as the foundation of the project.

- Generate documents such as agreements, contracts, NDAs, lease agreements and employment documents.
- Accept the essential information: parties involved, terms and conditions, and the effective date.
- Use Generative AI to produce customizable document content rather than a fixed, hard-coded template.
- Provide an editable preview so the user can correct or improve the text before using it.
- Export the final document as TXT, DOCX and PDF so it can be shared, printed or edited further.
- Support basic branding, such as a logo, headings and a footer, so output looks professional.

1.5 Proposed Solution in Brief

LegalEase is a web application in which the user selects a document type, enters the parties, terms, effective date and jurisdiction, and clicks a single button. The request travels to a backend service, which calls Google Gemini to create the document text. The generated draft returns to the interface where it can be previewed and edited, and then downloaded in the format of choice.

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User Input -> Streamlit UI -> FastAPI backend -> Gemini AI
   -> Generated Text -> Preview / Edit -> TXT | DOCX | PDF
```

1.6 Feature Prioritization

Not every idea could be delivered at once, so the team ranked the features using a simple must-have / should-have / could-have approach.

Feature	Priority	Reason
Document type selection and input form	Must have	Core input to the whole workflow
AI generation of structured content	Must have	The central value of the project
Preview of generated content	Must have	User must see the result before export
Editing before export	Should have	Drafts rarely need zero changes
TXT, DOCX and PDF export	Should have	Makes the draft usable outside the app
Logo, heading and footer branding	Could have	Improves appearance; not needed for core function

1.7 Feasibility Study

- **Technical feasibility:** All required components are available in Python – Streamlit for the interface, FastAPI for the API, the Gemini SDK for generation, and python-docx and FPDF for document output.
- **Economic feasibility:** The project relies on open-source libraries and a hosted generative AI service, so no specialized hardware or licensed software is required for a local demonstration.
- **Operational feasibility:** The interface is form-based and requires no legal or programming knowledge, so non-technical users can operate it.
- **Schedule feasibility:** The scope is limited to draft generation, preview, edit and export, which a four-member team can complete within a normal project cycle.

1.8 Risks and Considerations Identified

- **Accuracy of AI output:** generated text is a draft and may miss clauses or contain wording errors. The team decided the application must clearly state that drafts should be reviewed before real legal use.
- **Dependency on an external AI service:** generation needs a working API key and internet connection; failures must be handled gracefully.
- **Scope creep:** ideas such as user accounts or e-signatures were noted but deliberately left out of the initial scope.

1.9 Outcome of the Phase

The phase ended with an agreed project concept, a target-user list, a prioritized feature set and a shortlist of technologies. These outputs were handed to the Requirement Analysis phase, where the ideas were converted into formal functional and technical requirements.

Key decision: LegalEase generates editable drafts to assist the user. It does not replace professional legal review.